

TIC 467214801

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 0.760850 d (ALIASED, see below)
-> corrected period: 1.521701 d
Epoch (t0) 2825.3015 BTJD
Duration 0.0840 d
Depth 0.950 % (9,505 ppm)
SNR 2133.7

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

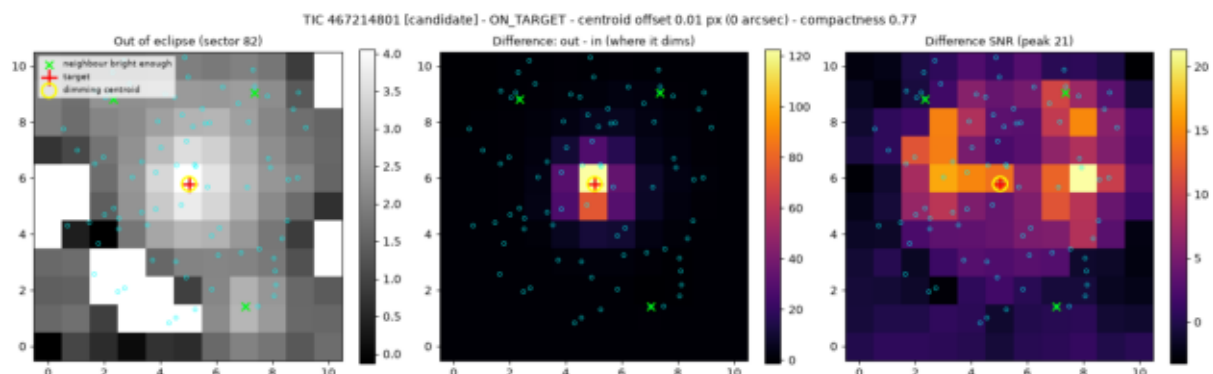
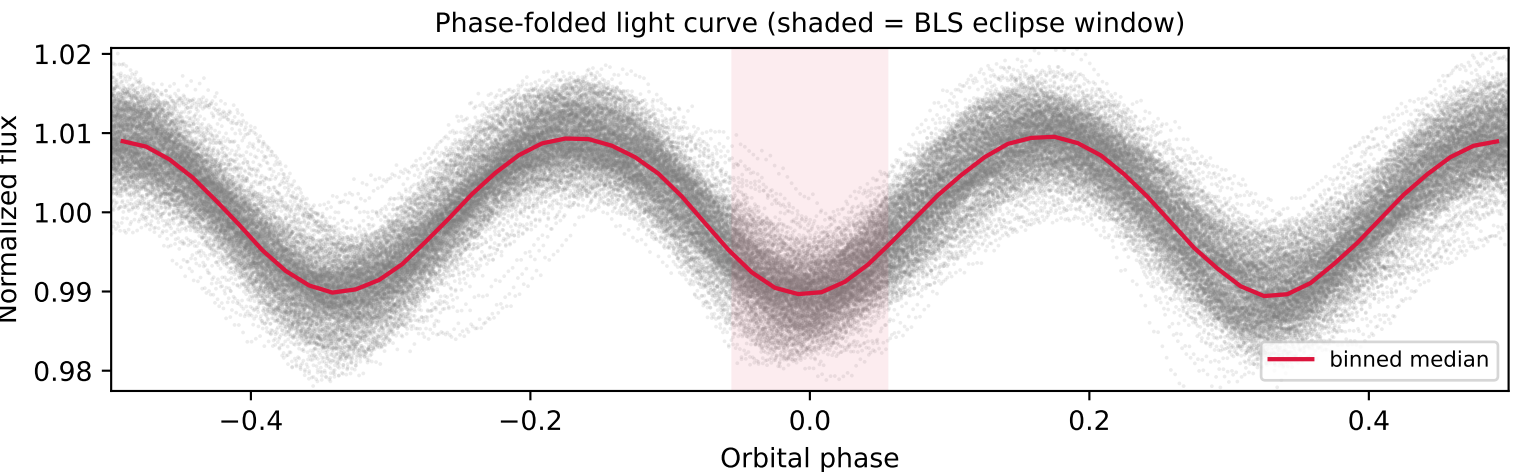
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 329 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 104.6 (gate ≥ 7.0)
Odd/even z-score 5.40 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 25.9 sigma t0 refined by -0.0100 phase
Difference-image peak SNR 21.36 (gate ≥ 5.0) Compactness 0.77 (gate ≥ 0.50)
Dimming centroid offset 0.01 px (0 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 467214771
4.01 px away, dTmag +4.53 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.521701 d, in any write-up.